ota-manager shadcn `:root`/`.light` palette — WCAG 2.1 contrast audit

Revision: 1 **Last modified:** 2026-07-09T18:51:55Z

Scope + why this audit exists

`clients/ota-manager` vendors the OpenDesign design-token system (`src/styles/opendesign-tokens.css`, imported first in `src/index.css`), but the shadcn/ui-generated HSL custom properties declared directly in `clients/ota-manager/src/index.css` under `:root` and `.light` are declared **after** the OpenDesign import and **win every one of the token-name collisions** (`--background`, `--foreground`, `--border`, `--ring`, etc. are defined by both layers under the same custom-property names, and CSS custom properties resolve to the last declaration in source order / highest specificity, which here is the shadcn block). Consequently the colors `clients/ota-manager` actually SHIPS to end users are the shadcn HSL triplets below, and OpenDesign's own contrast guarantees (if any) never take effect for this client. This audit computes real WCAG 2.1 contrast ratios for the shadcn palette that is actually rendered, so ota-manager's shipped colors have — for the first time — a captured, computed contrast record.

No component or CSS source file was modified to produce this audit. No git command was run. Every ratio quoted below is copy-pasted verbatim from the real, executed run of `contrast.py` (also in this directory) — nothing here is hand-computed or estimated (Constitution §11.4.6 no-guessing / §11.4.123 rock-solid-proof).

Theme-naming assumption (stated explicitly per §11.4.6)

`clients/ota-manager/src/index.css` defines the **dark** palette under the bare `:root` selector and the **light** palette under `.light`. There is `grep -n "^\.dark" src/index.css` → **no match**: no separate `.dark { }` block exists. `clients/ota-manager/src/stores/ui-store.ts` confirms this is intentional, not an oversight — its own code comment reads:

> “index.css base `:root` is the DARK palette and `@custom-variant dark (&:is(.dark *))` needs a `.dark` ancestor” — `applyThemeClass()` adds exactly one of the `light`/`dark` classes to `<html>`; when the class is `dark`, the bare `:root` values apply unmodified (CSS custom properties inherit down the tree, and a `.dark` selector with no properties of its own overrides nothing); when the class is `light`, the `.light` block’s declarations override `:root`. The store’s default `theme` is `"dark"`.

This audit therefore labels the `:root` block **“dark”** and the `.light` block **“light”** to match the ACTUAL rendered behaviour confirmed by the store code — not the common shadcn convention (where `:root` is usually the light theme and a `.dark` class supplies overrides). This is stated as an assumption per the task’s honest-boundary instruction, backed by the cited grep + source evidence, not guessed.

Token source (verbatim, read 2026-07-09)

`clients/ota-manager/src/index.css` lines 14–69 — see `contrast.py`’s `DARK_TOKENS` / `LIGHT_TOKENS` dicts for the exact literal copy consumed by the script.

Pairs audited + why each is real (usage evidence)

Every pair below is backed by a concrete component/file reference found by grepping `clients/ota-manager/src` — no pair was invented without seeing it consumed:

#	Pair	Class	Bar	Usage evidence
1	`foreground` / `background`	text	4.5:1	`index.css` body `{ color: hsl(var(--foreground)) }`

#	Pair	Class	Bar	Usage evidence
2	`foreground` / `card`	text	4.5:1	default text inside `card.tsx` content (inherits `foreground`)
3	`card-foreground` / `card`	text	4.5:1	`card.tsx` explicit `text-card-foreground` usages
4	`muted-foreground` / `background`	text	4.5:1	topbar/help text directly on page background
5	`muted-foreground` / `card`	text	4.5:1	secondary text in `data-table.tsx`, `dashboard-page.tsx`
6	`muted-foreground` / `sidebar-background`	text	4.5:1	`app-layout.tsx:79` “Helix OTA v1.0”, `:135` admin email
7	`primary-foreground` / `primary`	text	4.5:1	`button.tsx` default variant: `bg-primary text-primary-foreground`
8	`secondary-foreground` / `secondary`	text	4.5:1	`button.tsx` secondary variant
9	`accent-foreground` / `accent`	text	4.5:1	`button.tsx` outline/ghost hover state
10	`destructive-foreground` / `destructive`	text	4.5:1	`button.tsx` / `badge.tsx` destructive variant
11	`popover-foreground` / `popover`	text	4.5:1	`select.tsx` / `dialog.tsx` / `alert-dialog.tsx` content
12	`sidebar-foreground` / `sidebar-background`	text	4.5:1	`sidebar.tsx` / `app-layout.tsx` nav item, default state
13	`sidebar-foreground` / `sidebar-`	text	4.5:1	`app-layout.tsx:69` active/hover nav item — see note below

#	Pair	Class	Bar	Usage evidence
	`accent` (assumption)			
14	`border` / `background`	ui	3.0:1	`button.tsx` outline variant, `input.tsx`: `border border-input bg-background`
15	`border` / `card`	ui	3.0:1	card outer border against the card's own surface
16	`input` / `background`	ui	3.0:1	form-field border on page background
17	`ring` / `background`	ui	3.0:1	`button.tsx` `focus-visible:ring-1 focus-visible:ring-ring`
18	`ring` / `card`	ui	3.0:1	`badge.tsx` `focus:ring-2 focus:ring-ring` on card-surfaced controls
19	`sidebar-border` / `sidebar-background`	ui	3.0:1	`sidebar.tsx:38,84` section dividers

Note on pair #13 (stated assumption, §11.4.6): `app-layout.tsx:69` applies `[&.active]:bg-sidebar-accent [&.active]:text-sidebar-accent-foreground`, but `tailwind.config.ts`'s `sidebar` color group (lines 53–59) defines only a **flat** `accent: "hsl(var(--sidebar-accent))"` string — there is no nested `sidebar.accent.foreground` key, so the Tailwind JIT compiler has no color to resolve `text-sidebar-accent-foreground` against and that utility class does not exist in the generated stylesheet (verified: `grep -rn "sidebar-accent-foreground" src` finds it referenced only in `app-layout.tsx`, never defined as a resolvable color anywhere in `tailwind.config.ts`). This is a component/config wiring gap, not a token color-value issue, and is **out of this contrast audit's scope** (it was not edited, per the task's read-only constraint) — but it means the REAL rendered foreground color on an active/hover sidebar nav item is the inherited `sidebar-foreground` (the base class on that link, unaffected by the no-op hover/active utility), not a distinct "sidebar-accent-foreground" color. Pair #13 audits that real, rendered combination.

`contrast.py` — literal stdout (real run, not hand-transcribed)

Script: `contrast.py` in this same directory. Run as `python3 contrast.py`, Python 3.13.13, exit code 0. Full literal output (`contrast_output.txt` in this directory is the byte-identical captured file):

```
===================================================================
WCAG 2.1 contrast audit -- clients/ota-manager shadcn :root/.light
HSL tokens
SC 1.4.3 Contrast Minimum (text)      bar = 4.5:1
SC 1.4.11 Non-text Contrast (UI/large) bar = 3.0:1
===================================================================

--- THEME: dark (':root', default / '.dark'-equivalent) ---
[PASS] dark  | foreground / background                          |
fg=foreground(210 40% 98%) [#f8fafc] bg=background(222.2 84% 4.9%)
[#020817] | ratio=19.09:1  bar=4.5:1  class=text
[PASS] dark  | foreground / card                                |
fg=foreground(210 40% 98%) [#f8fafc] bg=card(222.2 84% 4.9%)
[#020817] | ratio=19.09:1  bar=4.5:1  class=text
[PASS] dark  | card-foreground / card                           |
fg=card-foreground(210 40% 98%) [#f8fafc] bg=card(222.2 84% 4.9%)
[#020817] | ratio=19.09:1  bar=4.5:1  class=text
[PASS] dark  | muted-foreground / background                    |
fg=muted-foreground(215 20.2% 65.1%) [#94a3b8] bg=background(222.2
84% 4.9%) [#020817] | ratio=7.80:1  bar=4.5:1  class=text
[PASS] dark  | muted-foreground / card                          |
fg=muted-foreground(215 20.2% 65.1%) [#94a3b8] bg=card(222.2 84%
4.9%) [#020817] | ratio=7.80:1  bar=4.5:1  class=text
[PASS] dark  | muted-foreground / sidebar-background            |
fg=muted-foreground(215 20.2% 65.1%) [#94a3b8] bg=sidebar-
background(222.2 84% 3%) [#01050e] | ratio=7.95:1  bar=4.5:1
class=text
[PASS] dark  | primary-foreground / primary                     |
fg=primary-foreground(222.2 47.4% 11.2%) [#0f172a]
bg=primary(217.2 91.2% 59.8%) [#3b82f6] | ratio=4.85:1  bar=4.5:1
class=text
[PASS] dark  | secondary-foreground / secondary                 |
fg=secondary-foreground(210 40% 98%) [#f8fafc] bg=secondary(217.2
32.6% 17.5%) [#1e293b] | ratio=13.95:1  bar=4.5:1  class=text
[PASS] dark  | accent-foreground / accent                       |
fg=accent-foreground(210 40% 98%) [#f8fafc] bg=accent(217.2 32.6%
```

```
17.5%) [#1e293b] | ratio=13.95:1  bar=4.5:1  class=text
[PASS] dark  | destructive-foreground / destructive            |
fg=destructive-foreground(210 40% 98%) [#f8fafc] bg=destructive(0
62.8% 30.6%) [#7f1d1d] | ratio=9.56:1  bar=4.5:1  class=text
[PASS] dark  | popover-foreground / popover                    |
fg=popover-foreground(210 40% 98%) [#f8fafc] bg=popover(222.2 84%
4.9%) [#020817] | ratio=19.09:1  bar=4.5:1  class=text
[PASS] dark  | sidebar-foreground / sidebar-background         |
fg=sidebar-foreground(210 40% 98%) [#f8fafc] bg=sidebar-
background(222.2 84% 3%) [#01050e] | ratio=19.47:1  bar=4.5:1
class=text
[PASS] dark  | sidebar-foreground / sidebar-accent (assumption) |
fg=sidebar-foreground(210 40% 98%) [#f8fafc] bg=sidebar-
accent(217.2 32.6% 14%) [#18212f] | ratio=15.46:1  bar=4.5:1
class=text
[FAIL] dark  | border / background                             |
fg=border(217.2 32.6% 17.5%) [#1e293b] bg=background(222.2 84%
4.9%) [#020817] | ratio=1.37:1  bar=3.0:1  class=ui
[FAIL] dark  | border / card                                   |
fg=border(217.2 32.6% 17.5%) [#1e293b] bg=card(222.2 84% 4.9%)
[#020817] | ratio=1.37:1  bar=3.0:1  class=ui
[FAIL] dark  | input / background                              |
fg=input(217.2 32.6% 17.5%) [#1e293b] bg=background(222.2 84%
4.9%) [#020817] | ratio=1.37:1  bar=3.0:1  class=ui
[FAIL] dark  | ring / background                               |
fg=ring(224.3 76.3% 48%) [#1d4ed8] bg=background(222.2 84% 4.9%)
[#020817] | ratio=2.98:1  bar=3.0:1  class=ui
[FAIL] dark  | ring / card                                     |
fg=ring(224.3 76.3% 48%) [#1d4ed8] bg=card(222.2 84% 4.9%)
[#020817] | ratio=2.98:1  bar=3.0:1  class=ui
[FAIL] dark  | sidebar-border / sidebar-background             |
fg=sidebar-border(217.2 32.6% 14%) [#18212f] bg=sidebar-
background(222.2 84% 3%) [#01050e] | ratio=1.26:1  bar=3.0:1
class=ui

--- THEME: light ('.light' class) ---
[PASS] light | foreground / background                         |
fg=foreground(222.2 84% 4.9%) [#020817] bg=background(0 0% 100%)
[#ffffff] | ratio=19.99:1  bar=4.5:1  class=text
[PASS] light | foreground / card                               |
fg=foreground(222.2 84% 4.9%) [#020817] bg=card(0 0% 100%)
[#ffffff] | ratio=19.99:1  bar=4.5:1  class=text
[PASS] light | card-foreground / card                          |
fg=card-foreground(222.2 84% 4.9%) [#020817] bg=card(0 0% 100%)
```

```
[#ffffff] | ratio=19.99:1  bar=4.5:1  class=text
[PASS] light | muted-foreground / background                         |
fg=muted-foreground(215.4 16.3% 46.9%) [#64748b] bg=background(0
0% 100%) [#ffffff] | ratio=4.75:1  bar=4.5:1  class=text
[PASS] light | muted-foreground / card                               |
fg=muted-foreground(215.4 16.3% 46.9%) [#64748b] bg=card(0 0%
100%) [#ffffff] | ratio=4.75:1  bar=4.5:1  class=text
[PASS] light | muted-foreground / sidebar-background                 |
fg=muted-foreground(215.4 16.3% 46.9%) [#64748b] bg=sidebar-
background(210 40% 98%) [#f8fafc] | ratio=4.54:1  bar=4.5:1
class=text
[PASS] light | primary-foreground / primary                          |
fg=primary-foreground(210 40% 98%) [#f8fafc] bg=primary(221.2
83.2% 53.3%) [#2563eb] | ratio=4.94:1  bar=4.5:1  class=text
[PASS] light | secondary-foreground / secondary                      |
fg=secondary-foreground(222.2 47.4% 11.2%) [#0f172a]
bg=secondary(210 40% 96.1%) [#f1f5f9] | ratio=16.30:1  bar=4.5:1
class=text
[PASS] light | accent-foreground / accent                            |
fg=accent-foreground(222.2 47.4% 11.2%) [#0f172a] bg=accent(210
40% 96.1%) [#f1f5f9] | ratio=16.30:1  bar=4.5:1  class=text
[PASS] light | destructive-foreground / destructive                  |
fg=destructive-foreground(210 40% 98%) [#f8fafc] bg=destructive(0
72.2% 50.6%) [#dc2626] | ratio=4.61:1  bar=4.5:1  class=text
[PASS] light | popover-foreground / popover                          |
fg=popover-foreground(222.2 84% 4.9%) [#020817] bg=popover(0 0%
100%) [#ffffff] | ratio=19.99:1  bar=4.5:1  class=text
[PASS] light | sidebar-foreground / sidebar-background               |
fg=sidebar-foreground(222.2 84% 4.9%) [#020817] bg=sidebar-
background(210 40% 98%) [#f8fafc] | ratio=19.09:1  bar=4.5:1
class=text
[PASS] light | sidebar-foreground / sidebar-accent (assumption) |
fg=sidebar-foreground(222.2 84% 4.9%) [#020817] bg=sidebar-
accent(210 40% 90%) [#dbe6f0] | ratio=15.74:1  bar=4.5:1
class=text
[FAIL] light | border / background                                   |
fg=border(214.3 31.8% 91.4%) [#e2e8f0] bg=background(0 0% 100%)
[#ffffff] | ratio=1.23:1  bar=3.0:1  class=ui
[FAIL] light | border / card                                         |
fg=border(214.3 31.8% 91.4%) [#e2e8f0] bg=card(0 0% 100%)
[#ffffff] | ratio=1.23:1  bar=3.0:1  class=ui
[FAIL] light | input / background                                    |
fg=input(214.3 31.8% 91.4%) [#e2e8f0] bg=background(0 0% 100%)
[#ffffff] | ratio=1.23:1  bar=3.0:1  class=ui
```

```
[PASS] light | ring / background                               |
fg=ring(221.2 83.2% 53.3%) [#2563eb] bg=background(0 0% 100%)
[#ffffff] | ratio=5.17:1  bar=3.0:1  class=ui
[PASS] light | ring / card                                     |
fg=ring(221.2 83.2% 53.3%) [#2563eb] bg=card(0 0% 100%) [#ffffff]
| ratio=5.17:1  bar=3.0:1  class=ui
[FAIL] light | sidebar-border / sidebar-background             |
fg=sidebar-border(214.3 31.8% 88%) [#d7dfea] bg=sidebar-
background(210 40% 98%) [#f8fafc] | ratio=1.28:1  bar=3.0:1
class=ui

===================================================================
SUMMARY: 38 pairs checked, 28 PASS, 10 FAIL
===================================================================

FAILURES + proposed same-hue-family AA-passing tone (foreground
lightness adjusted only):

* theme=dark pair='border / background'
    current: --border: 217.2 32.6% 17.5%;   ratio=1.37:1  (bar
3.0:1)  FAIL
    proposed: --border: 217.2 32.6000% 39.5434%;   recomputed
ratio=3.00:1  PASS
    proposed hex: #445d86 (was #1e293b)

* theme=dark pair='border / card'
    current: --border: 217.2 32.6% 17.5%;   ratio=1.37:1  (bar
3.0:1)  FAIL
    proposed: --border: 217.2 32.6000% 39.5434%;   recomputed
ratio=3.00:1  PASS
    proposed hex: #445d86 (was #1e293b)

* theme=dark pair='input / background'
    current: --input: 217.2 32.6% 17.5%;   ratio=1.37:1  (bar
3.0:1)  FAIL
    proposed: --input: 217.2 32.6000% 39.5434%;   recomputed
ratio=3.00:1  PASS
    proposed hex: #445d86 (was #1e293b)

* theme=dark pair='ring / background'
    current: --ring: 224.3 76.3% 48%;   ratio=2.98:1  (bar 3.0:1)
FAIL
    proposed: --ring: 224.3 76.3000% 48.2524%;   recomputed
ratio=3.00:1  PASS
```

```
    proposed hex: #1d4ed9 (was #1d4ed8)

* theme=dark pair='ring / card'
    current: --ring: 224.3 76.3% 48%;   ratio=2.98:1  (bar 3.0:1)
FAIL
    proposed: --ring: 224.3 76.3000% 48.2524%;   recomputed
ratio=3.00:1  PASS
    proposed hex: #1d4ed9 (was #1d4ed8)

* theme=dark pair='sidebar-border / sidebar-background'
    current: --sidebar-border: 217.2 32.6% 14%;   ratio=1.26:1
(bar 3.0:1)  FAIL
    proposed: --sidebar-border: 217.2 32.6000% 38.9887%;
recomputed ratio=3.00:1  PASS
    proposed hex: #435c84 (was #18212f)

* theme=light pair='border / background'
    current: --border: 214.3 31.8% 91.4%;   ratio=1.23:1  (bar
3.0:1)  FAIL
    proposed: --border: 214.3 31.8000% 60.9885%;   recomputed
ratio=3.00:1  PASS
    proposed hex: #7c97bb (was #e2e8f0)

* theme=light pair='border / card'
    current: --border: 214.3 31.8% 91.4%;   ratio=1.23:1  (bar
3.0:1)  FAIL
    proposed: --border: 214.3 31.8000% 60.9885%;   recomputed
ratio=3.00:1  PASS
    proposed hex: #7c97bb (was #e2e8f0)

* theme=light pair='input / background'
    current: --input: 214.3 31.8% 91.4%;   ratio=1.23:1  (bar
3.0:1)  FAIL
    proposed: --input: 214.3 31.8000% 60.9885%;   recomputed
ratio=3.00:1  PASS
    proposed hex: #7c97bb (was #e2e8f0)

* theme=light pair='sidebar-border / sidebar-background'
    current: --sidebar-border: 214.3 31.8% 88%;   ratio=1.28:1
(bar 3.0:1)  FAIL
    proposed: --sidebar-border: 214.3 31.8000% 59.6316%;
recomputed ratio=3.00:1  PASS
    proposed hex: #7793b9 (was #d7dfea)
```

Bug found + fixed while building this audit (honest disclosure, §11.4.1/§11.4.102)

The first draft of `contrast.py`'s `propose_fix()` auto-fix search had two real bugs, found and fixed before this report was finalized (not hand-waved away):

1. **Wrong-endpoint check.** The darkening search direction (`direction=-1`, searching toward lightness `0.0`) incorrectly checked `ratio_at(hi)` — which for that direction is `orig_l`, the ALREADY-FAILING starting lightness — instead of `ratio_at(lo)` (the `l=0.0` extreme it was actually searching toward). This caused the darkening direction to be skipped even when it was the correct fix (e.g. all four `light`-theme failures below need a DARKER border, not a lighter one, since the light theme's borders already sit on a near-white background). Fixed by checking the ratio at the correct extreme for each direction.
2. **Unguarded sentinel string.** When no reachable lightness cleared the bar, the function returned a human-readable `"UNCONFIRMED: ..."` string in the slot meant for an HSL token, and the report code unconditionally passed it to `hsl_to_hex()`, which crashed trying to `.split()` it into three HSL fields. Fixed by returning `None` for the unreachable case and guarding the print path to report `UNCONFIRMED` honestly instead of crashing or fabricating a hex value.

Both fixes are in the `contrast.py` committed in this directory; the stdout above is from the POST-fix run (exit code 0, no traceback).

AA failures — summary + proposed fixes

All 10 failures are **UI-component boundary pairs (SC 1.4.11, 3.0:1 bar)** — zero text pairs (SC 1.4.3) fail in either theme. Every failure is a “subtle divider” color (`border`/`input`/`sidebar-border`) or the focus `ring` sitting only marginally under the 3.0:1 bar:

Theme	Token pair	Ratio	Bar	Proposed token (same hue+sat)	New ratio
dark	`border` / `background`	1.37:1	3.0:1	`217.2 32.6%`	3.00:1

Theme	Token pair	Ratio	Bar	Proposed token (same hue+sat)	New ratio
				`39.5%` (`#445d86`)	
dark	`border` / `card`	1.37:1	3.0:1	`217.2 32.6% 39.5%` (`#445d86`)	3.00:1
dark	`input` / `background`	1.37:1	3.0:1	`217.2 32.6% 39.5%` (`#445d86`)	3.00:1
dark	`ring` / `background`	2.98:1	3.0:1	`224.3 76.3% 48.25%` (`#1d4ed9`)	3.00:1
dark	`ring` / `card`	2.98:1	3.0:1	`224.3 76.3% 48.25%` (`#1d4ed9`)	3.00:1
dark	`sidebar-border` / `sidebar-background`	1.26:1	3.0:1	`217.2 32.6% 39.0%` (`#435c84`)	3.00:1
light	`border` / `background`	1.23:1	3.0:1	`214.3 31.8% 61.0%` (`#7c97bb`)	3.00:1
light	`border` / `card`	1.23:1	3.0:1	`214.3 31.8% 61.0%` (`#7c97bb`)	3.00:1
light	`input` / `background`	1.23:1	3.0:1	`214.3 31.8% 61.0%` (`#7c97bb`)	3.00:1
light	`sidebar-border` /	1.28:1	3.0:1	`214.3 31.8%`	3.00:1

Theme	Token pair	Ratio	Bar	Proposed token (same hue+sat)	New ratio
	`sidebar-background`			`59.6% (#7793b9)`	

Notes on the proposed values:

- Each proposed lightness is the exact output of `propose_fix()`'s binary search targeting `ratio == bar` (3.00:1) to the nearest reachable value, keeping hue+saturation fixed (minimal, same-hue-family shift, per the task instruction) — not a hand-picked round number. A project adopting these would likely round to a slightly higher ratio (e.g. 3.1–3.2:1) for headroom against rounding/rendering variance; the exact binary-search target is reported here for traceability.
- `border`, `input`, and (in the dark theme) `card`'s border context share one proposed value per theme because `--border` and `--input` are numerically **identical** tokens in this codebase in both themes (verified from the literal `index.css` values: dark `--border: 217.2 32.6% 17.5%` == `--input: 217.2 32.6% 17.5%`; light `--border: 214.3 31.8% 91.4%` == `--input: 214.3 31.8% 91.4%`) — this is a genuine property of the current token set, not an audit artefact.
- `--ring`'s dark-theme failure is marginal (2.98:1 vs 3.00:1 bar — a 0.02:1 shortfall) and disappears with a ~0.25-percentage-point lightness bump; light theme's `ring` already PASSes (5.17:1).

Honest boundary on SC 1.4.11 applicability (§11.4.6)

SC 1.4.11 (Non-text Contrast) requires 3:1 contrast for "the visual information required to identify user interface components and states," except for inactive/disabled components, or where the appearance is "determined by the user agent and not modified by the author." Two of the failing pairs deserve an explicit scope note rather than a flat verdict:

- `border` / `card` — a card's outer border is frequently REDUNDANT with layout spacing/shadow that already delineates the card boundary; if `card.tsx`'s surrounding layout provides that redundant cue, this specific pair may not be a hard SC 1.4.11 violation in practice, only a best-practice gap. This report does NOT

assert component-level redundancy either way (that requires inspecting rendered layout, out of this read-only token audit's scope) — it reports the raw token-pair ratio and flags this nuance rather than asserting a black-and-white violation.
- `ring` (focus indicator) and `border`/`input` on actual form controls (`input.tsx`) are NOT redundant — the focus ring is frequently the ONLY visual cue of keyboard focus, and an input's border is frequently the only cue of the field's clickable/editable boundary. These are the pairs SC 1.4.11's own normative examples call out directly, so these two are reported as unambiguous failures.

No ratio anywhere in this document was guessed; every number is the literal computed output of `contrast.py`, pasted verbatim above.

Sources verified

- WCAG 2.1 Success Criterion 1.4.3 Contrast (Minimum): https://www.w3.org/WAI/WCAG21/Understanding/contrast-minimum.html
- WCAG 2.1 Success Criterion 1.4.11 Non-text Contrast: https://www.w3.org/WAI/WCAG21/Understanding/non-text-contrast.html

(Sources verified 2026-07-09 against the canonical W3C "Understanding WCAG 2.1" pages for these two success criteria; the 4.5:1 / 3.0:1 bars and the `(L1+0.05)/(L2+0.05)` relative-luminance contrast formula implemented in `contrast.py` match these normative definitions.)